

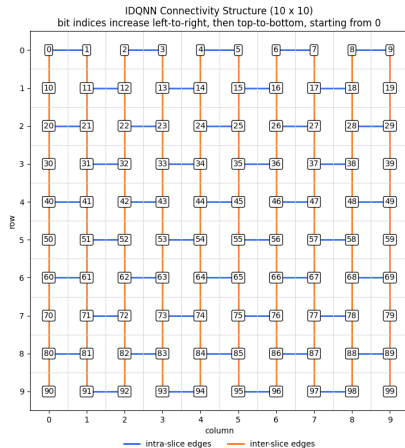
Why IDQNN Fails on Handwritten Digits

A First Diagnostic Draft

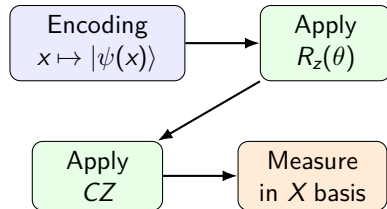
Quantum Machine Learning

April 24, 2026

IDQNN Structure



Nearest-neighbor CZ structure on the 10×10 bit grid.



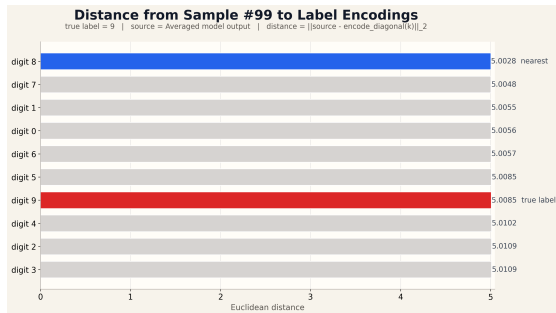
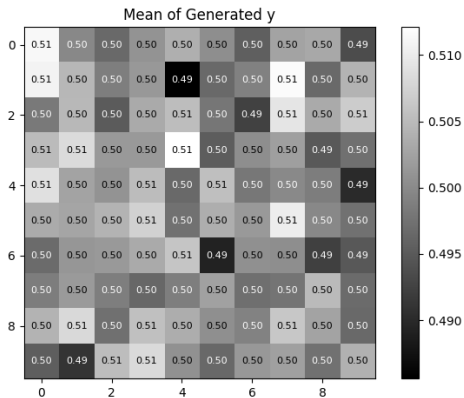
$$|\psi_{\text{out}}\rangle = (\prod CZ)(\prod R_z(\theta))|\psi(x)\rangle, \quad y \sim p_\theta(y|x)$$

White pixels(1) are encoded to $|0\rangle$, black pixels(0) to $|+\rangle$.

Generate the corresponding y bits by measuring the qubits.

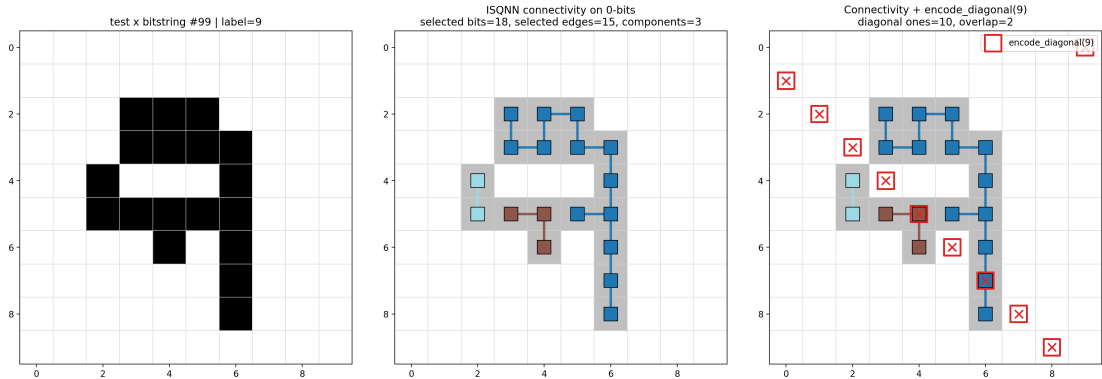
Test Sample #99: Distance between Output and Label-Diagonal Encoding

10000 test-time samples for #99 are generated and averaged.



The nearest label for sample #99 is digit 8 rather than the true label 9.

Test Sample #99: output distribution structure



- Measurement outcomes of White pixels are always 50/50 .
- overlap with the label-diagonal encoding is very limited.

Outputs distributions for black pixels blocks:

101

$$|0\rangle|+\rangle|0\rangle$$

$$|0\rangle \left(|+\rangle \cos \frac{\theta_2}{2} - i|-\rangle \sin \frac{\theta_2}{2} \right) |0\rangle.$$

1001

$$|0\rangle|+\rangle|+\rangle|0\rangle$$

$$P(m_2, m_3) = \frac{1}{4} (1 + m_2 m_3 \sin \theta_2 \sin \theta_3)$$

$$P(++) = P(--)= \frac{1}{4} (1 + \sin \theta_2 \sin \theta_3)$$

$$P(+-) = P(-+) = \frac{1}{4} (1 - \sin \theta_2 \sin \theta_3)$$

10001

$$|0\rangle|+\rangle|+\rangle|+\rangle|0\rangle$$

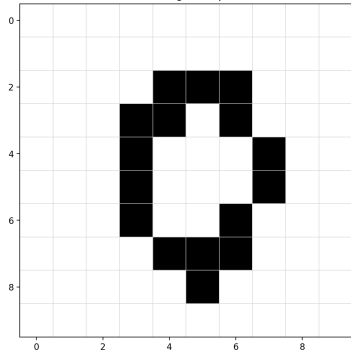
$$P(m_2, m_3, m_4) = \frac{1}{8} [1 + m_2 m_4 \cos \theta_2 \cos \theta_4 - m_2 m_3 m_4 \cos \theta_3 \sin \theta_2 \sin \theta_4]$$

$$m_j = \begin{cases} +1, & |+\rangle \\ -1, & |-\rangle \end{cases}$$

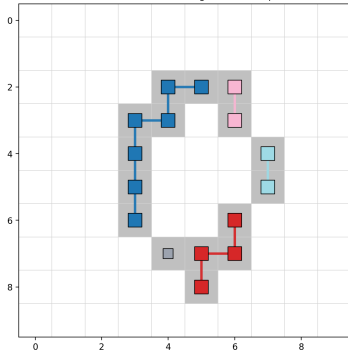
- 101: one tunable internal bit.
- 1001: pairwise correlation appears.
- 10001: higher-order but still local correlation.
- Outer qubits measurements outcomes remain 50/50 in the X basis.

Connectivity Example: Test Sample #157

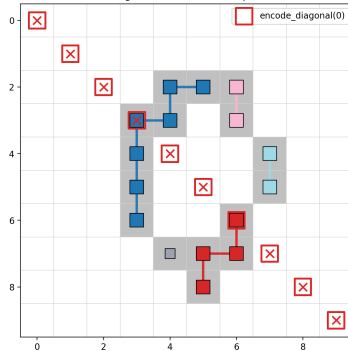
test x bitstring #157 | label=0



ISQNN connectivity on 0-bits
selected bits=16, selected edges=11, components=5

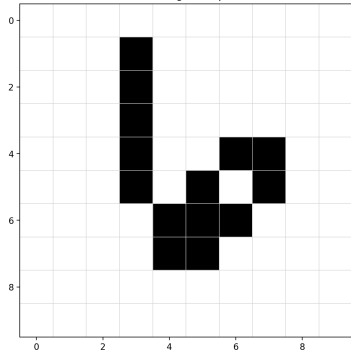


Connectivity + encode_diagonal(0)
diagonal ones=10, overlap=2

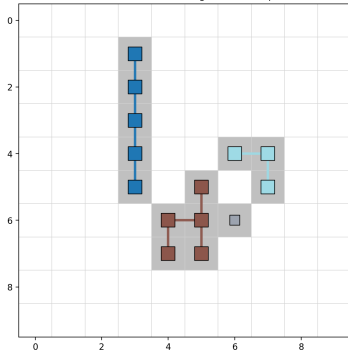


Connectivity Example: Test Sample #164

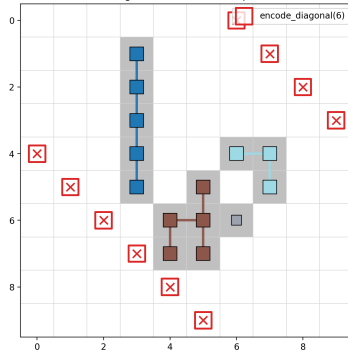
test x bitstring #164 | label=6



ISQNN connectivity on 0-bits
selected bits=14, selected edges=10, components=4



Connectivity + encode_diagonal(6)
diagonal ones=10, overlap=0



Factorized Output Family

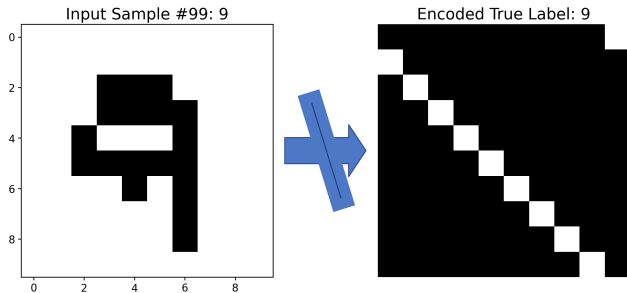
For a fixed input pattern x :

$$P(y \mid x, \theta) = 2^{-|\mathcal{W}(x)|} \prod_{C \in \mathcal{C}(x)} P_C(y_C \mid \theta_C),$$

where $\mathcal{W}(x)$ are pixels encoded to $|0\rangle$ and $\mathcal{C}(x)$ are CZ-connected components on pixels encoded to $|+\rangle$.

- White pixels contribute only uniform 50/50 outputs in the X basis.
- Nontrivial correlations exist only inside connected black-pixel blocks.
- Different blocks multiply independently.

Implication for Encoded Labels



- The pre-encoded label is a global target pattern, but the IDQNN output for a handwritten-digit input is only a product of local connected-block distributions and uniform white-pixel noise.

Structural Limitation

Therefore IDQNN is structurally unable to reproduce any global pre-encoded label as a generic inference target for MNIST handwritten digits.